

- 6 (a) Two similar coils **A** and **B** of insulated wire are wound on to a soft-iron core, as illustrated in Fig. 6.1.

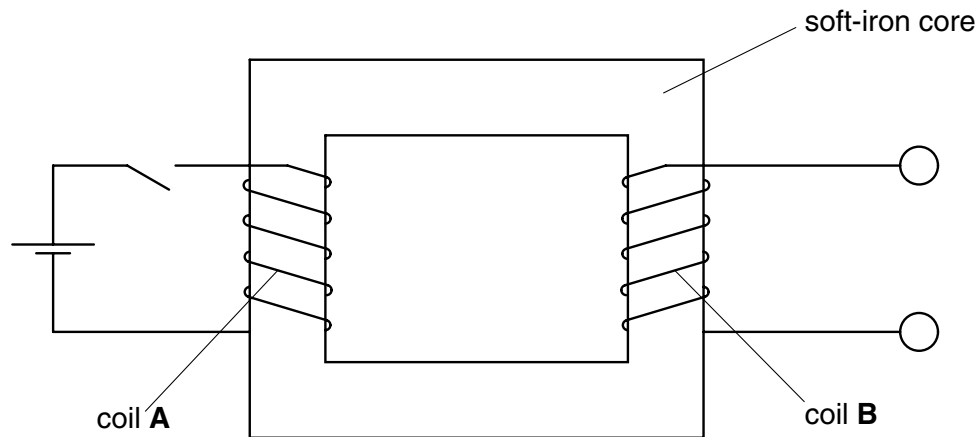


Fig. 6.1

When the current I in coil **A** is switched on and then off, the variation with time t of the current is shown in Fig. 6.2.

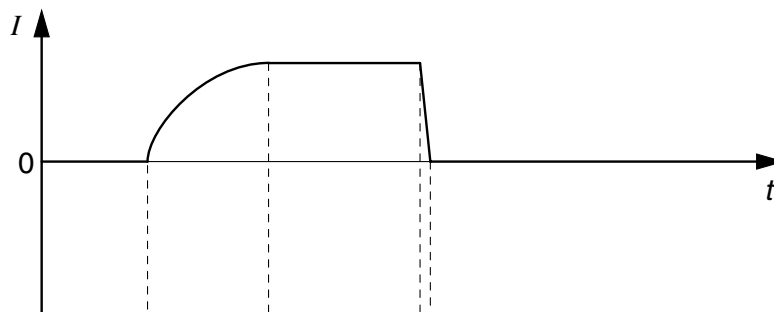


Fig. 6.2

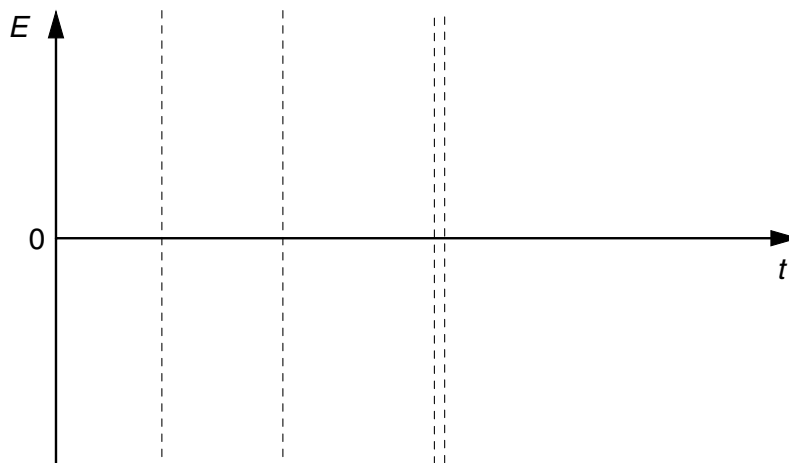


Fig. 6.3

On Fig. 6.3, draw a graph to show the variation with time t of the e.m.f. E induced in coil **B**.

[3]

(b) Fig. 6.4 is the circuit of a bridge rectifier.

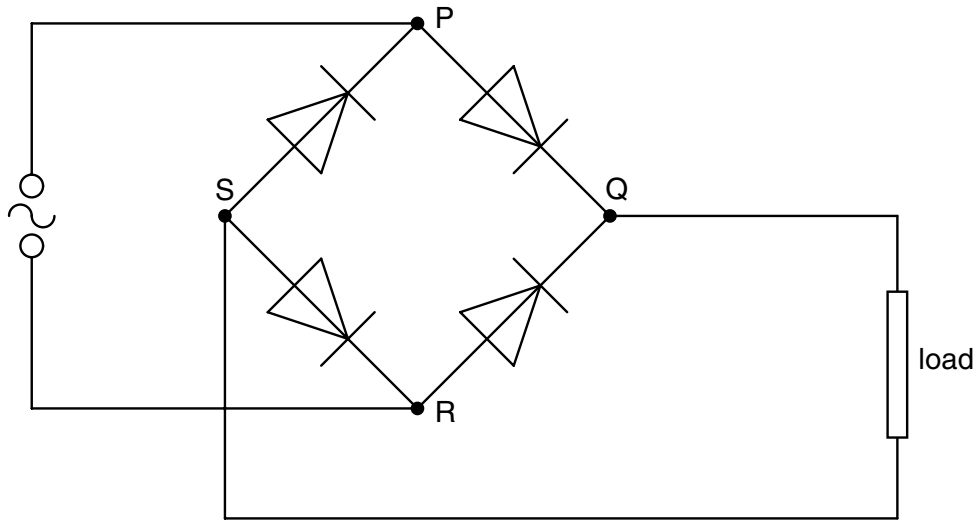


Fig. 6.4

An alternating supply connected across PR has an output of 6.0 V r.m.s.

- (i) On Fig. 6.4, circle those diodes that are conducting when R is positive with respect to P. [1]
- (ii) Calculate the maximum potential difference between points Q and S, assuming that the diodes are ideal.

potential difference = V [2]

- (iii) State and explain how a capacitor may be used to smooth the output from the rectifier. You may draw on Fig. 6.4 if you wish.

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[3]